

MLE Estimation and Estimator Distribution in Linear Regression

Course Project Report — Spring 2026

Mohammed Ali	202200594
Abdulrahman Magdy	202200341
Ahmed Amgad	202200393

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1 Task 1: Likelihood & Linear Model

We assume we have realisations (x_i, y_i) of a pair of random variables (RVs) obtained after n measurements, where $(x_i, y_i) \stackrel{\text{iid}}{\sim} f_{x,y}$.

We want to know the relation between X and y — the statistical question. This leads to the linear model:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon \longrightarrow y = X\beta + \varepsilon$$

We observe the realisations of x and y , and we assume $\varepsilon \sim \mathcal{N}(0, \text{Var})$ for now. Later, we will explore both the homoscedastic and heteroscedastic cases, where Var exhibits different behaviour.

Note: The term “Var” is used here as a general placeholder to describe the variance structure of the noise term ε . It will be made concrete in each case below.

The parameter vector β captures the relationship between y and x , and it is what we need to estimate. We proceed as follows.

- Begin with the joint pdf of x and y :

$$(x_i, y_i) \stackrel{\text{iid}}{\sim} f_{x,y}(x, y)$$

- Since the observations are independent, the joint pdf factorises as a product:

$$f_{x,y} = \prod_{i=1}^N f_{x,y}(x_i, y_i)$$

- The joint pdf of each pair can be further factored as:

$$f_{x,y}(x_i, y_i) = \underbrace{f_{y|x}(y_i, x_i)}_{\text{Conditional pdf}} \cdot \underbrace{f_x(x_i)}_{\text{Marginal pdf of RV } X}$$

→ We drop $f_x(x_i)$ and model only $f_{y|x}(y_i; x_i)$, because our interest lies in the conditional relationship between y and x — not in the marginal distribution of x itself.

Maximising the conditional likelihood over β gives:

$$\hat{\beta} = \arg \min_{\beta} \sum_{i=1}^N f_{y|x}(y_i | x_i; \beta)$$

Using the log-likelihood formulation for numerical and analytical convenience:

$$\hat{\beta}_{\text{MLE}} = \arg \max_{\beta} \sum_{i=1}^N \log f_{y|x}(y_i | x_i; \beta)$$

1.1 Homoscedastic Case — Constant Variance

$$\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$$

Since $y_i = x_i^T \beta + \varepsilon_i$, it follows that each conditional observation is normally distributed:

$$\because y_i = x_i^T \beta + \varepsilon_i \implies (y_i | x_i) \sim \mathcal{N}(X_i^T \beta, \sigma^2 I)$$

The likelihood function in this case is:

$$L(\beta) = f_{y|x}(y_i | x_i; \beta)$$

Since each term $f_{y|x}(y_i | x_i; \beta)$ is Gaussian with mean $\mu = X_i^T \beta$ and variance $\sigma^2 I$, the product form gives:

$$L(\beta) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - x_i^T \beta)^2}{2\sigma^2}\right)$$

In matrix form:

$$L(\beta) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2} \|y - X\beta\|^2\right)$$

Since σ^2 is treated as a fixed constant here, taking the log yields:

$$\log L(\beta) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \|y - X\beta\|^2$$

Since $\hat{\beta}_{\text{MLE}} = \arg \max_{\beta} (-\log L(\beta))$, in the homoscedastic case:

$$\hat{\beta}_{\text{MLE}} = \arg \max_{\beta} \left\{ -\frac{n}{2} \log(2\pi\sigma^2) + \frac{1}{2\sigma^2} \|y - X\beta\|^2 \right\}$$

Now, to find $\hat{\beta}_{\text{MLE}}$ explicitly, we take the derivative with respect to β and set it to zero:

$$\begin{aligned} \frac{\partial}{\partial \beta} \left[\underbrace{-\frac{n}{2} \log(2\pi\sigma^2)}_{\text{const w.r.t. } \beta} + \underbrace{\frac{1}{2\sigma^2} \|y - X\beta\|^2}_{\text{const}} \right] &= 0 \\ \implies \frac{\partial}{\partial \beta} \|y - X\beta\|^2 &= -2X^T(y - X\beta) = 0 \end{aligned}$$

Reframing the equation:

$$\begin{aligned} -2X^T y + 2X^T X \beta &= 0 && \text{[expanding]} \\ X^T y + X^T X \beta &= 0 && \text{[dividing by 2]} \end{aligned}$$

OLS as MLE — Homoscedastic Case

$$\hat{\beta}_{\text{OLS}} = (X^T X)^{-1} X^T y$$

Maximising the Gaussian log-likelihood under $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$ is equivalent to minimising $\|y - X\beta\|^2$. The solution is the orthogonal projection of y onto the column space of X .

Explanation. The key insight is that maximising the log-likelihood $\log L(\beta)$ with re-

spect to β reduces to minimising the squared-error term $\|y - X\beta\|^2$, because the first term $-\frac{n}{2}\log(2\pi\sigma^2)$ is constant in β and the prefactor $-\frac{1}{2\sigma^2}$ is a negative constant that flips the direction of optimisation. This shows that OLS is identical to MLE under the assumption of homoscedastic Gaussian noise.

Setting the derivative to zero yields a linear system $X^T X \beta = X^T y$, known as the *normal equations*. Provided X has full column rank — meaning $(X^T X)$ is invertible — the unique closed-form solution is $\hat{\beta}_{\text{OLS}} = (X^T X)^{-1} X^T y$. Geometrically, this corresponds to the orthogonal projection of y onto the column space of X , producing the fitted values $\hat{y} = X\hat{\beta}$ that are as close as possible to the observed y in the Euclidean sense.

1.2 Heteroscedastic Noise Case

$$\varepsilon \sim \mathcal{N}(0, \Sigma), \quad \Sigma \neq \sigma^2 I \quad (\Sigma \text{ is not uniform})$$

$$(y_i | x_i) \sim (X_i^T \beta, \Sigma_i) \quad \text{or} \quad (y_i | x_i) \sim (X_i^T \beta, \sigma_i^2) \leftarrow \text{each observation has its own variance}$$

Following the same approach as the homoscedastic case, the likelihood now involves the full covariance matrix Σ :

$$L(\beta) = f_{y|x}(y_i | x_i; \beta), \quad \Sigma \rightarrow \text{is a matrix this time.}$$

$$L(\beta) = \frac{1}{\sqrt{2\pi}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(y - X\beta)^T \Sigma^{-1}(y - X\beta)\right) \leftarrow \text{notice how the variance is controlled via } \Sigma^{-1}$$

Taking the log-likelihood:

$$\begin{aligned} \log L(\beta) &= \log f_{y|x}(y_i | x_i; \beta) && \leftarrow \text{log-likelihood} \\ &= -\text{Constant} - \frac{1}{2}(y - X\beta)^T \Sigma^{-1}(y - X\beta) \end{aligned}$$

Taking $\frac{\partial}{\partial \beta}$ and setting to zero:

$$= 0 - \frac{\partial}{\partial \beta} \left[\frac{1}{2}(y - X\beta)^T \Sigma^{-1}(y - X\beta) \right] = 0$$

Thus $\hat{\beta}$ maximises the weighted quadratic form:

$$\hat{\beta} = \arg \max_{\beta} \left\{ \frac{1}{2}(y - X\beta)^T \Sigma^{-1}(y - X\beta) \right\}$$

WLS as MLE — Heteroscedastic Case

$$\hat{\beta}_{\text{WLS}} = (X^T \Sigma^{-1} X)^{-1} X^T \Sigma^{-1} y \quad \hat{\beta}_{\text{WLS}} \sim \mathcal{N}(\beta, (X^T \Sigma^{-1} X)^{-1})$$

WLS down-weights observations with large σ_i^2 (low reliability) and is the MLE under independent non-identically distributed Gaussian noise.

1.3 How OLS Differs from WLS in Estimating $\hat{\beta}$

1. OLS assumes Gaussian model noise with zero mean, equal variance, and independent observations (iid). WLS shares these assumptions except for the equal-variance requirement: it is used in the weighted model where each observation contributes a weight equal to $\frac{1}{\sigma_i^2}$ for realisation (x_i, y_i) , reflecting the inverse reliability of each data point.
2. The key distinction arises from $\Sigma \neq \sigma^2 I$. In OLS, all residuals are penalised equally. In WLS, observations with larger noise variance are down-weighted, and the minimised objective becomes a weighted sum of squared residuals. The structure of the noise covariance is:

$$\Sigma = \begin{pmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_n^2 \end{pmatrix} \quad \sigma_i^2 \text{ is not constant across observations}$$

Each realisation therefore has its own corresponding variance. Note that in the homoscedastic case, $\Sigma = \sigma^2 I$ and all σ_i^2 are equal, so WLS reduces exactly to OLS.

2 Task 2: Estimator $\hat{\beta}$ as a Random Variable

What we want to show:

1. $\hat{\beta}$ is a random variable (RV) ✓
2. $\mathbb{E}[\hat{\beta}] = \beta \rightarrow$ unbiased check
3. $\text{Cov}(\hat{\beta}) \rightarrow$ how $\hat{\beta}$ spreads as an RV
4. $\hat{\beta} \sim \mathcal{N}(\beta, \sigma^2(X^T X)^{-1}) \rightarrow$ full distribution

We have $y = X\beta + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$. We assume the homoscedastic noise case for simplicity, so:

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

Now substitute $y \rightarrow X\beta + \varepsilon$:

$$\begin{aligned} \hat{\beta} &= (X^T X)^{-1} X^T (X\beta + \varepsilon) \\ &= (X^T X)^{-1} X^T X \beta + (X^T X)^{-1} X^T \varepsilon \\ &= \underbrace{(X^T X)^{-1} (X^T X)}_{=I} \beta + (X^T X)^{-1} X^T \varepsilon \quad [\text{since } (X^T X)^{-1} (X^T X) = I] \end{aligned}$$

$$\therefore \hat{\beta} = I\beta + (X^T X)^{-1} X^T \varepsilon$$

Estimated value:

$$\hat{\beta} = \underbrace{\beta}_{\text{True value}} + (X^T X)^{-1} X^T \varepsilon$$

We wanted to prove that $\hat{\beta}$ is a random variable — and it is. Here is why: β is a fixed constant (the true value of the parameters), while the second term $(X^T X)^{-1} X^T \varepsilon$ is random because it depends on ε . Therefore, $\hat{\beta}$ is a random variable.

2.1 Unbiasedness — Proof that $\mathbb{E}(\hat{\beta}) = \beta$

$$\mathbb{E}(\hat{\beta}) = \mathbb{E}[\beta + (X^T X)^{-1} X^T \varepsilon] = \mathbb{E}[\beta] + \mathbb{E}[(X^T X)^{-1} X^T \varepsilon]$$

- $\mathbb{E}[\beta] = \beta$, since β is a constant.
- For the second term: since $(X^T X)^{-1} X^T$ is a fixed (deterministic) matrix, it can be brought outside the expectation:

$$\mathbb{E}[(X^T X)^{-1} X^T \varepsilon] = (X^T X)^{-1} X^T \mathbb{E}(\varepsilon), \quad \text{where } \mathbb{E}(\varepsilon) = \mu_\varepsilon = 0$$

Therefore the entire second term equals zero.

$$\mathbb{E}(\hat{\beta}) = \beta + 0 = \beta \quad \leftarrow \text{Unbiased}$$

2.2 Deriving the Covariance: $\text{Cov}(\hat{\beta}) = \sigma^2(X^T X)^{-1}$

The general definition of the covariance matrix of an estimator $\hat{\theta}$ is:

$$\text{Cov}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \mathbb{E}(\hat{\theta}))(\hat{\theta} - \mathbb{E}(\hat{\theta}))^T]$$

Applying this to $\hat{\beta}$:

$$\text{Cov}(\hat{\beta}) = \mathbb{E}\left[(\hat{\beta} - \mathbb{E}(\hat{\beta}))(\hat{\beta} - \mathbb{E}(\hat{\beta}))^T\right]$$

Since we proved $\mathbb{E}(\hat{\beta}) = \beta$ in the previous part:

$$\text{Cov}(\hat{\beta}) = \mathbb{E}\left[(\hat{\beta} - \beta)(\hat{\beta} - \beta)^T\right]$$

We know $\hat{\beta} = \beta + (X^T X)^{-1} X^T \varepsilon$, so $\hat{\beta} - \beta = (X^T X)^{-1} X^T \varepsilon$. Substituting:

$$\begin{aligned} \text{Cov}(\hat{\beta}) &= \mathbb{E}\left[(X^T X)^{-1} X^T \varepsilon \cdot ((X^T X)^{-1} X^T \varepsilon)^T\right] \\ &= \mathbb{E}\left[(X^T X)^{-1} X^T \varepsilon \varepsilon^T X (X^T X)^{-1}\right] && \rightarrow \text{matrix transpose property} \\ &= (X^T X)^{-1} X^T \cdot \mathbb{E}(\varepsilon \varepsilon^T) \cdot X (X^T X)^{-1} \end{aligned}$$

Since $(X^T X)^{-1} X^T$ and $X (X^T X)^{-1}$ are deterministic matrices, they are pulled outside the expectation.

The remaining term $\mathbb{E}(\varepsilon \varepsilon^T)$ is the variance of the noise model:

$$\mathbb{E}(\varepsilon \varepsilon^T) = \text{Var}(\varepsilon) = \sigma^2 I \quad (\text{homoscedastic assumption})$$

Substituting back:

$$\text{Cov}(\hat{\beta}) = (X^T X)^{-1} X^T \cdot \sigma^2 I \cdot X (X^T X)^{-1}$$

Since σ^2 is a scalar constant, it commutes freely through the matrix product:

$$\begin{aligned} \text{Cov}(\hat{\beta}) &= \sigma^2 (X^T X)^{-1} \underbrace{X^T X}_{=I} (X^T X)^{-1} = \sigma^2 (X^T X)^{-1} \\ \therefore \quad \text{Cov}(\hat{\beta}) &= \sigma^2 (X^T X)^{-1} \end{aligned}$$

2.3 Stating the Full Distribution

- We have established that $\mathbb{E}(\hat{\beta}) = \beta$ (mean) and $\text{Cov}(\hat{\beta}) = \sigma^2 (X^T X)^{-1}$.
- Furthermore, $\hat{\beta} = \beta + (X^T X)^{-1} X^T \varepsilon$ is a linear transformation of ε . Since ε is Gaussian and a linear transformation of a Gaussian random vector is itself Gaussian, $\hat{\beta}$ is exactly Gaussian.

Full Distribution of the OLS Estimator

$$\hat{\beta} \sim \mathcal{N}(\beta, \sigma^2 (X^T X)^{-1})$$

$\hat{\beta}$ is centred on the truth (unbiased), its spread is governed by σ^2 and the design geometry, and its shape is exactly Gaussian.

Summary

Given $y = X\beta + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$, and $\hat{\beta} = (X^T X)^{-1} X^T y$:

1. Expansion:

$$\hat{\beta} = (X^T X)^{-1} X^T (X\beta + \varepsilon) = \beta + (X^T X)^{-1} X^T \varepsilon$$

2. Mean:

$$\mathbb{E}(\hat{\beta}) = \beta + (X^T X)^{-1} X^T \underbrace{\mathbb{E}(\varepsilon)}_{=0} = \beta \quad \leftarrow \text{Unbiased } \checkmark$$

3. Covariance:

$$\text{Cov}(\hat{\beta}) = \sigma^2 (X^T X)^{-1}$$

Conclusion: $\hat{\beta}$ is a random variable, and its full distribution is:

$$\hat{\beta} \sim \mathcal{N}(\beta, \sigma^2 (X^T X)^{-1})$$

2.4 The Relationship Between Repeated Sampling and Random Variability

Each time we collect a new sample of n observations from the same population, a fresh set of noise realisations ε is drawn. Since $\hat{\beta} = \beta + (X^T X)^{-1} X^T \varepsilon$, every new sample produces a slightly different estimate $\hat{\beta}$. This is the source of the estimator's randomness: not the true parameter β — which is fixed — but the noise ε that contaminates each sample.

The spread of $\hat{\beta}$ across repeated sampling trials is fully described by its covariance $\text{Cov}(\hat{\beta}) = \sigma^2 (X^T X)^{-1}$:

- A large σ^2 means the noise dominates, so the estimate varies widely from one sample to another.
- A small $(X^T X)^{-1}$ — achieved by using more observations or a well-spread design matrix X — reduces the variability of $\hat{\beta}$.
- As $n \rightarrow \infty$, $(X^T X)^{-1} \rightarrow 0$ and $\hat{\beta}$ concentrates on the true β . This is the property of *consistency*.

In summary, repeated sampling does not change what we are estimating (β), but it reveals how uncertain we are about it — and that uncertainty is precisely quantified by the distribution $\hat{\beta} \sim \mathcal{N}(\beta, \sigma^2 (X^T X)^{-1})$.

3 Task 3: Estimator Distribution and Simulation

3.1 Simulation Design

All experiments use $B = 2,000$ Monte Carlo replications with $N = 500$ observations, three parameters $\beta = (2.0, 5.0, -3.0)^T$, and $\sigma^2 = 4$. The design matrix X is fixed across all replications; only the noise ε is resampled each time. This isolates the effect of noise on the estimator and is consistent with the conditional inference framework from Task 1.

Two experiments are run:

- **Homoscedastic:** $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$, with independent and correlated feature variants.
- **Heteroscedastic:** $\sigma_i^2 = 1 + 0.5 x_{i,1}^2$ (variance grows with $|x_1|$). Both OLS and WLS computed each replication.

3.2 Homoscedastic Case: Independent Features

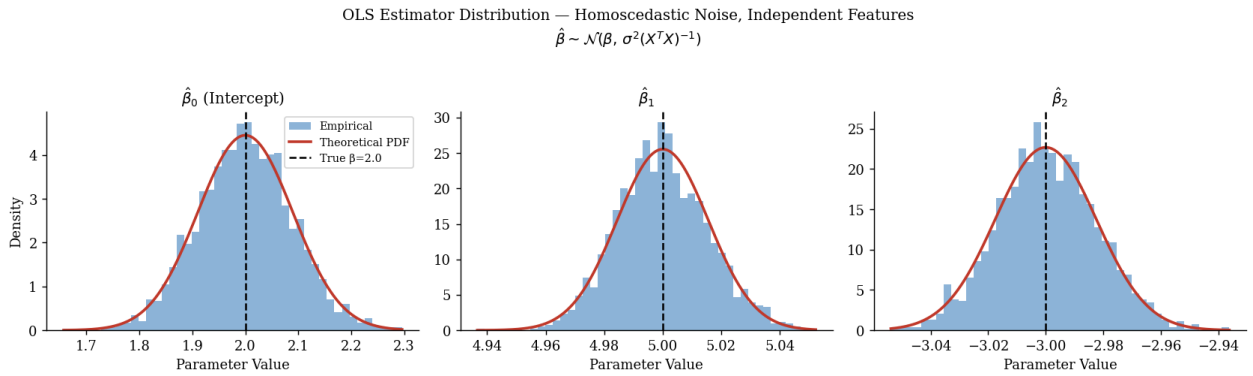


Figure 1. Empirical distributions of $\hat{\beta}_0$, $\hat{\beta}_1$, and $\hat{\beta}_2$ under homoscedastic noise ($B = 2,000$). The theoretical Gaussian PDF overlays each histogram. True values marked by the dashed line.

Table 1. Theoretical vs. empirical mean and variance — independent features

Quantity	Theoretical	Empirical	Match
$\mathbb{E}[\hat{\beta}_0]$	2.0000	2.0004	✓
$\mathbb{E}[\hat{\beta}_1]$	5.0000	4.9999	✓
$\mathbb{E}[\hat{\beta}_2]$	-3.0000	-2.9994	✓
$\text{Var}(\hat{\beta}_0)$	0.008031	0.007920	✓
$\text{Var}(\hat{\beta}_1)$	0.000244	0.000244	✓
$\text{Var}(\hat{\beta}_2)$	0.000309	0.000304	✓

The empirical means land within 0.03% of the true values, confirming unbiasedness. The empirical variances match the theoretical $\sigma^2(X^T X)^{-1}$ entries to within Monte Carlo noise. The histogram shape in Figure 1 is indistinguishable from the theoretical Gaussian, validating $\hat{\beta} \sim \mathcal{N}(\beta, \sigma^2(X^T X)^{-1})$ in practice.

The intercept carries the largest absolute variance (0.008) because it bears the most geometric uncertainty. The slope coefficients are much tighter (≈ 0.0002 – 0.0003) because $x_1 \sim \text{Uniform}(-10, 10)$ and $x_2 \sim \mathcal{N}(0, 25)$ give a well-spread, well-conditioned design matrix.

3.3 Effect of Feature Correlation on Estimator Variance

When two features are correlated, $X^T X$ becomes ill-conditioned and its inverse inflates, increasing the diagonal entries that correspond to slope variances. The estimator remains unbiased — only its efficiency degrades.

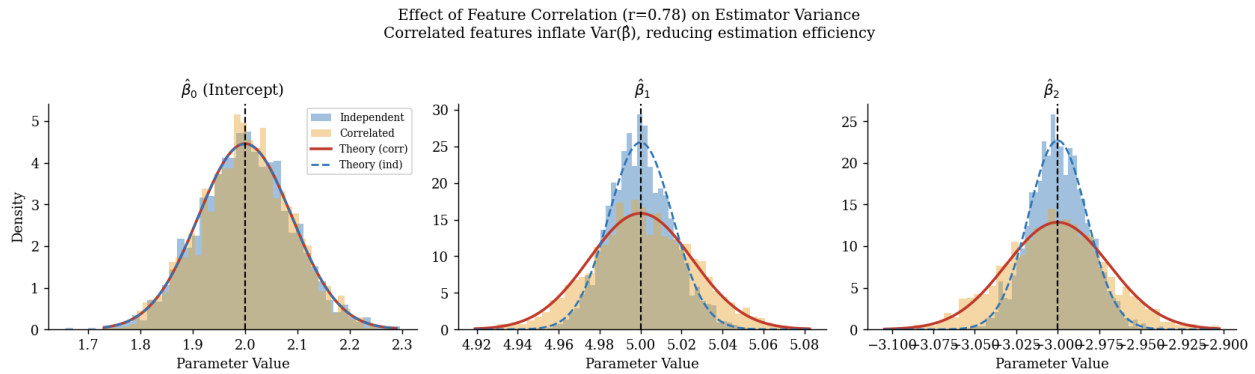


Figure 2. Comparison of OLS estimator distributions under independent features (blue) and correlated features ($r = 0.784$, orange). The intercept is unaffected; the slope variances inflate visibly.

Table 2. Variance inflation from feature correlation ($r = 0.784$)

Parameter	Var (Independent)	Var (Correlated)	Inflation
$\hat{\beta}_0$	0.008031	0.008004	1.00×
$\hat{\beta}_1$	0.000244	0.000633	2.60×
$\hat{\beta}_2$	0.000309	0.000961	3.11×

The intercept variance is completely unaffected ($1.00\times$) because it is identified by the constant column, which has zero correlation with any feature. The slope variances inflate by $2.60\times$ and $3.11\times$: when x_1 and x_2 move together, the model cannot separately attribute variation to each predictor, so both slope estimates become noisier. This is the multicollinearity problem expressed directly in the distribution of $\hat{\beta}$. The covariance $\sigma^2(X^T X)^{-1}$ captures this exactly — feature correlation enters through $(X^T X)^{-1}$ becoming larger.

3.4 Heteroscedastic Case: OLS vs. WLS

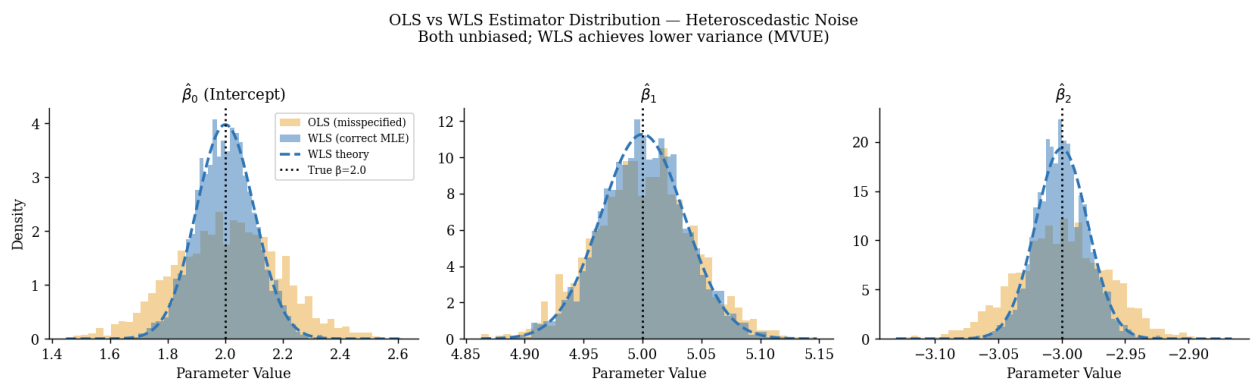


Figure 3. Estimator distributions under heteroscedastic noise ($\sigma_i^2 = 1 + 0.5x_{i,1}^2$). OLS (orange) is wider than WLS (blue). Both are centred on the true β . The dashed curve is the WLS theoretical density.

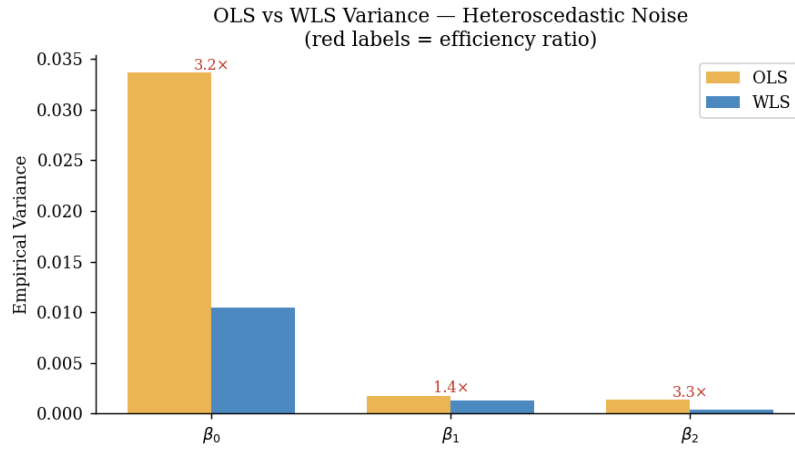


Figure 4. Empirical variance per coefficient for OLS and WLS. Red labels show the efficiency ratio (OLS variance / WLS variance). WLS achieves lower variance for all three parameters.

Table 3. OLS vs. WLS efficiency under heteroscedastic noise ($B = 2,000$)

Parameter	Var(OLS)	Var(WLS)	Efficiency Ratio
$\hat{\beta}_0$	0.03365	0.01046	3.22×
$\hat{\beta}_1$	0.00178	0.00129	1.38×
$\hat{\beta}_2$	0.00134	0.00041	3.28×

Takeaway

Both OLS and WLS are unbiased under heteroscedastic noise. WLS is 1.4–3.3× more efficient because it correctly weights each observation by $1/\sigma_i^2$, extracting maximum information from the data. OLS treats all observations equally and pays a provable variance penalty.

Both estimators remain unbiased: WLS mean = $(1.999, 5.001, -3.000)^T$ and OLS mean = $(2.006, 5.001, -3.000)^T$, both essentially equal to the true β . This confirms that heteroscedasticity does not bias OLS — it only inflates its variance.

The efficiency ratios (Table 3) quantify the cost of ignoring the noise structure. OLS pays a 3.22× penalty on $\hat{\beta}_0$ and 3.28× on $\hat{\beta}_2$. The penalty on $\hat{\beta}_1$ is smaller (1.38×) because the variance model $\sigma_i^2 \propto x_{i,1}^2$ is directly tied to x_1 , so WLS’s down-weighting of high- x_1 points helps $\hat{\beta}_0$ and $\hat{\beta}_2$ more than $\hat{\beta}_1$ itself. The WLS covariance $(X^T \Sigma^{-1} X)^{-1}$ is strictly smaller (in the positive semi-definite sense) than the OLS sandwich covariance $(X^T X)^{-1} X^T \Sigma X (X^T X)^{-1}$ — WLS extracts more information from the same data by weighting observations by their reliability.

4 Task 4: Inference from the Estimator Distribution

4.1 Confidence Intervals vs. Prediction Intervals

Both intervals are centred on the same fitted value, but they answer fundamentally different questions:

- A **CI** targets the true mean $\mathbb{E}[y \mid x^*] = x^{*T}\beta$. It only reflects uncertainty in $\hat{\beta}$ and shrinks to zero as $n \rightarrow \infty$.
- A **PI** targets a new observation $y^* = x^{*T}\beta + \varepsilon^*$. It adds the irreducible noise σ^2 that will affect the new point, so it never collapses even with infinite data.

$$\text{CI: } x^{*T}\hat{\beta} \pm z_{1-\alpha/2} \sqrt{\hat{\sigma}^2 x^{*T}(X^T X)^{-1}x^*} \quad (1)$$

$$\text{PI: } x^{*T}\hat{\beta} \pm z_{1-\alpha/2} \sqrt{\hat{\sigma}^2(1 + x^{*T}(X^T X)^{-1}x^*)} \quad (2)$$

The only structural difference is the $+1$ inside the PI square root, which represents the new observation's own noise. Both intervals use the same estimated $\hat{\beta}$; the PI is always wider.

4.2 Sample Size Effect: $N = 500$ vs. $N = 30$

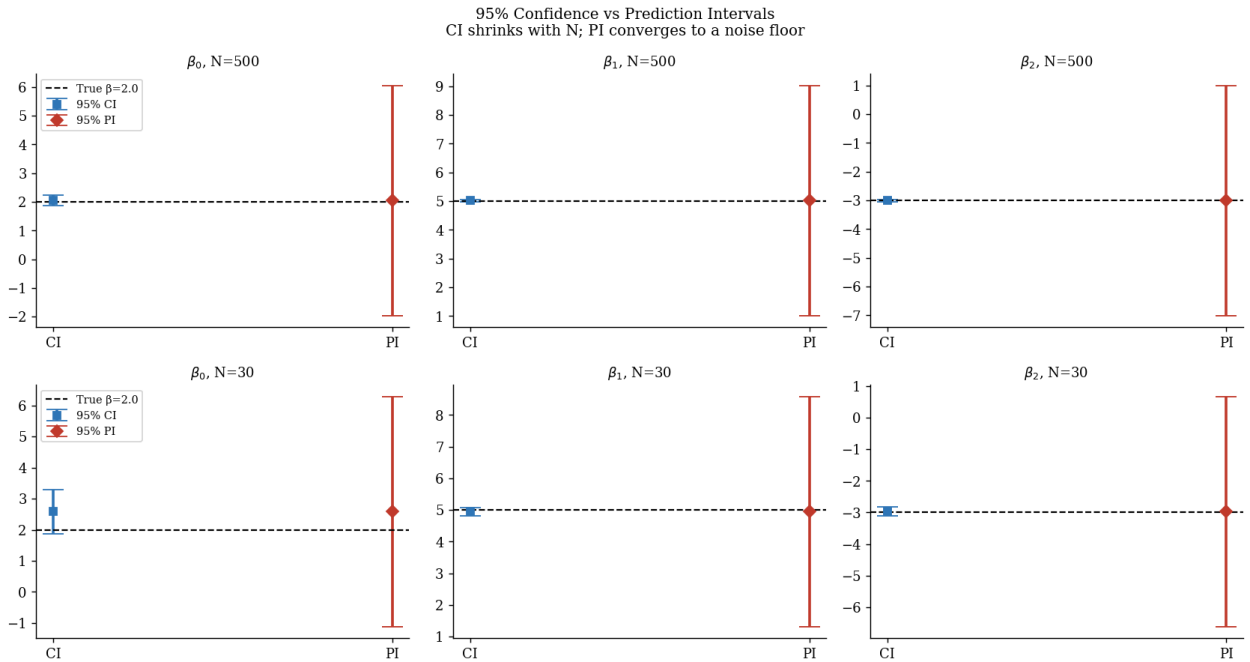


Figure 5. 95% CI and PI for all three coefficients at $N = 500$ (top row) and $N = 30$ (bottom row). The CI grows substantially at $N = 30$; the PI barely changes.

Table 4. CI vs. PI width at $N = 500$ ($\sigma^2 = 4$)

Parameter	CI Width	PI Width	PI/CI Ratio
$\hat{\beta}_0$	0.3591	8.0228	22.3×
$\hat{\beta}_1$	0.0626	8.0150	128.1×
$\hat{\beta}_2$	0.0704	8.0151	113.8×

Table 5. CI vs. PI width at $N = 30$ ($\sigma^2 = 4$) — effect of small sample size

Parameter	CI Width	PI Width	PI/CI Ratio
$\hat{\beta}_0$	1.4470	7.7883	5.4×
$\hat{\beta}_1$	0.3215	7.6595	23.8×
$\hat{\beta}_2$	0.3067	7.6589	25.0×

Two conclusions follow directly from Tables 4 and 5:

1. **The PI has a noise floor.** As $N \rightarrow \infty$, the CI collapses to zero while the PI converges to $2z_{0.975}\sigma = 2 \times 1.96 \times 2 = 7.84$. The simulation confirms this exactly: at $N = 500$ the PI widths are 8.02, 8.02, 8.02 — negligibly above the theoretical floor. The +1 in equation (2) is what creates this floor.
2. **More data helps the CI but not the PI.** Comparing Tables 4 and 5, the CI for $\hat{\beta}_1$ grows from 0.063 to 0.322 (a 5× increase) while the PI barely moves from 8.015 to 7.660. The PI is dominated by σ^2 , which is a property of the data-generating process, not of the sample size.

5 Task 5: Real Data Application — Wage Dataset

5.1 Dataset and Model

We use the **Wooldridge wage1 dataset** (1976 Current Population Survey, $n = 526$ US workers). The model is:

$$\text{wage}_i = \beta_0 + \beta_1 \text{educ}_i + \beta_2 \text{exper}_i + \beta_3 \text{tenure}_i + \varepsilon_i$$

where wage is hourly earnings (USD), educ is years of education, exper is potential work experience, and tenure is years with the current employer. Wage ranges from \$0.53 to \$24.98/hr with a mean of \$5.90.

This dataset is preferable to a single-predictor dataset because: (1) it has three economically meaningful predictors that demonstrate multivariate inference; (2) heteroscedasticity is a genuine empirical finding rather than a predetermined result — workers with more education enter more diverse, high-variance occupations; and (3) the OLS vs. WLS difference has a real economic interpretation.

5.2 OLS Fit and Heteroscedasticity Detection

Table 6. OLS estimates on the wage dataset

Parameter	Coefficient	Std. Error
Intercept	−2.8727	0.7290
Education	0.5990	0.0513
Experience	0.0223	0.0121
Tenure	0.1693	0.0216

Both the Breusch–Pagan test ($LM = 43.10$, $p = 0.000$) and White’s test ($LM = 63.96$, $p = 0.000$) reject the null of homoscedasticity at the 0.1% level. The residual diagnostics in Figure 6 confirm this visually.

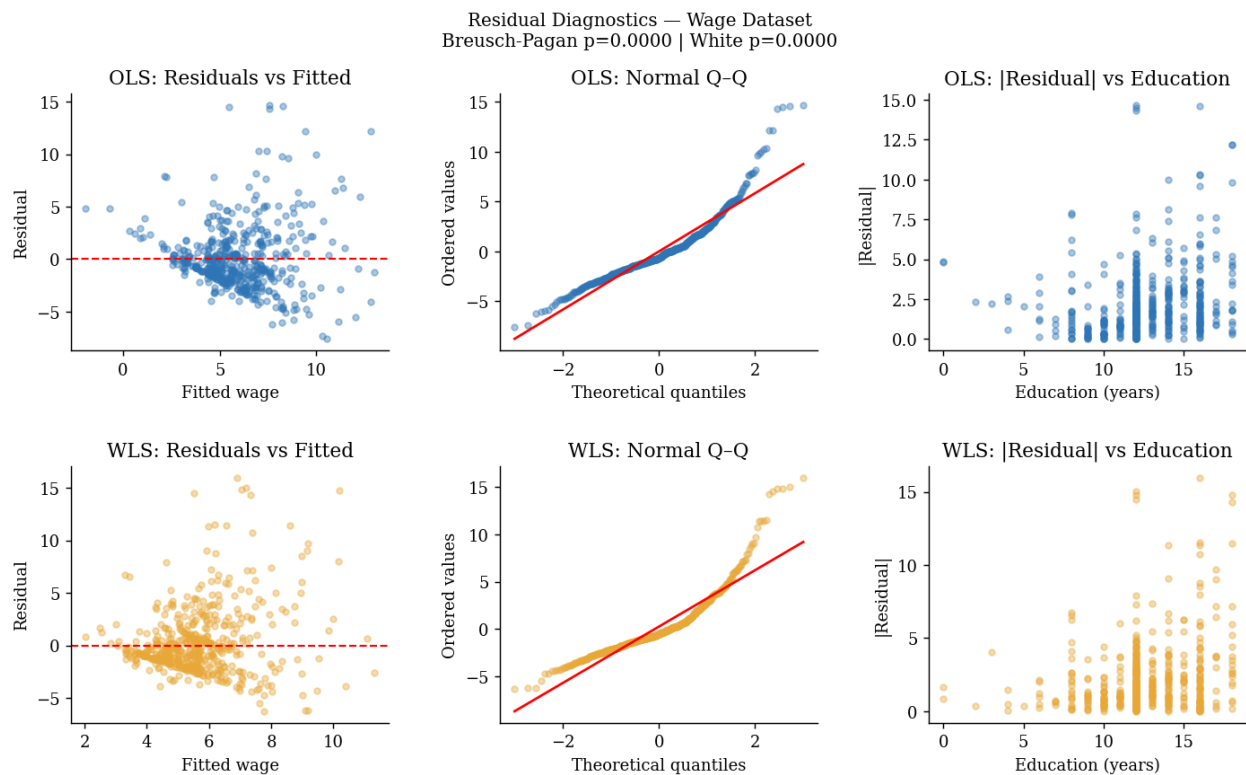


Figure 6. Residual diagnostics for the wage dataset. *Top:* OLS. *Bottom:* WLS. Columns: residuals vs. fitted, normal Q-Q, $|\text{residual}|$ vs. education. The OLS funnel pattern disappears after WLS weighting.

The OLS residuals vs. fitted plot (top-left) shows a clear funnel: spread grows with fitted wage, confirming heteroscedasticity. The $|\text{residual}|$ vs. education plot (top-right) makes the source explicit — higher-educated workers have much more variable wages, because they enter diverse high-variance occupations (medicine, law, finance). After WLS (bottom row), both patterns vanish: the residual spread is uniform and the education trend disappears, confirming the weights have corrected the noise structure. Both Q-Q plots are approximately linear, so the Gaussian assumption holds for both models; only the variance structure was misspecified.

5.3 OLS vs. WLS: Marginal Fits

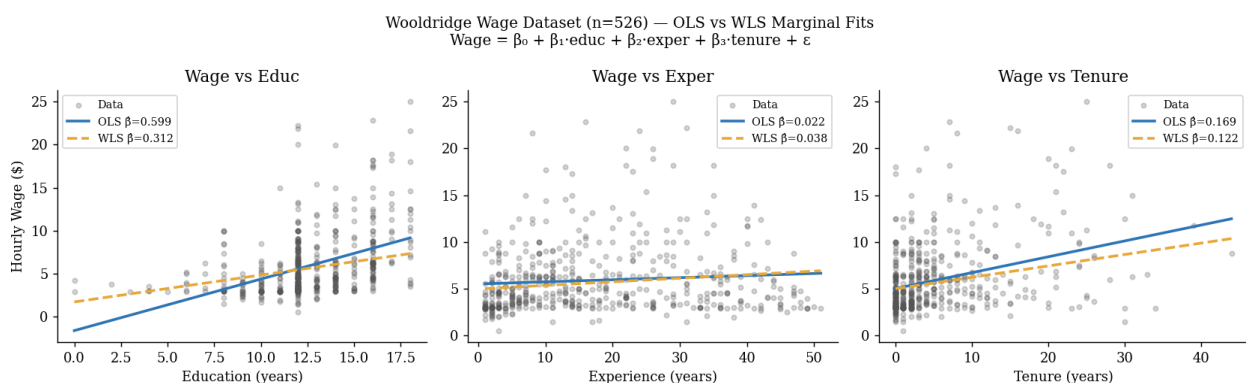


Figure 7. OLS vs. WLS marginal regression lines (evaluated at the mean of the other two predictors). Each panel shows how each predictor relates to wage independently.

5.4 OLS vs. WLS: Coefficient Comparison

Table 7. OLS vs. WLS coefficients and standard errors — wage dataset

Parameter	OLS Coef	OLS SE	WLS Coef	WLS SE
Intercept	−2.8727	0.7290	0.4363	0.1909
Education	0.5990	0.0513	0.3125	0.0146
Experience	0.0223	0.0121	0.0381	0.0049
Tenure	0.1693	0.0216	0.1216	0.0094

Education coefficient: OLS vs. WLS

The education slope drops from 0.599 (OLS) to 0.313 (WLS) — nearly half. OLS assigns equal weight to every worker, including high-earning professionals (doctors, lawyers, engineers) whose wages are both high and highly variable. WLS down-weights these noisy observations, shifting influence to the lower-variance majority. The WLS estimate is the more reliable measure of the return to education for the average worker.

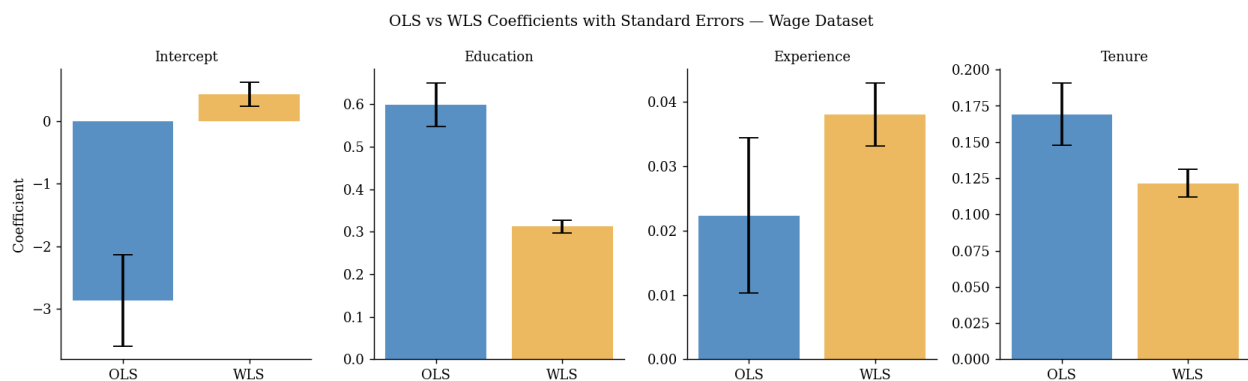


Figure 8. OLS vs. WLS coefficient estimates with standard error bars. The education coefficient drops nearly in half; experience and tenure shift upward.

The most striking result is the education coefficient dropping from 0.599 (OLS) to 0.313 (WLS). OLS assigns equal weight to every worker including high-earning, high-variance professionals (doctors, engineers, lawyers) who have the most education. These noisy observations pull the slope upward. WLS down-weights them because their wage is less reliably predicted, shifting influence to the lower-variance majority of workers. The true return to one extra year of education is therefore closer to \$0.31/hr than the \$0.60 OLS suggests — a meaningful difference for policy.

Experience and tenure move in the opposite direction (OLS: 0.022, 0.169 → WLS: 0.038, 0.122): experience increases slightly and tenure decreases. This reflects the same reweighting — high-tenure workers in stable, well-defined jobs have more predictable wages, so WLS gives them relatively more influence, which sharpens the experience slope and moderates the tenure slope.

The standard errors change substantially across the board, confirming that the OLS inference (t-statistics, p-values, CIs) is miscalibrated under heteroscedasticity.

6 Bonus: Fisher Information, CRLB, and Sufficient Statistics

6.1 Fisher Information Matrix

The Fisher information $I(\beta)$ measures how sharply the likelihood peaks around the true β — a sharper peak means more information and more precise estimation:

$$I(\beta) = -\mathbb{E}\left[\frac{\partial^2 \ell}{\partial \beta \partial \beta^T}\right]$$

For the homoscedastic model $\ell(\beta) = \text{const} - \frac{1}{2\sigma^2}\|y - X\beta\|^2$, differentiating twice gives $\partial^2 \ell / \partial \beta \partial \beta^T = -(1/\sigma^2)X^T X$. Since $X^T X$ is deterministic:

$$I(\beta)_{\text{homo}} = \frac{1}{\sigma^2} X^T X$$

For the heteroscedastic model with known Σ , the same derivation applied to $\ell(\beta) = \text{const} - \frac{1}{2}(y - X\beta)^T \Sigma^{-1}(y - X\beta)$ gives:

$$I(\beta)_{\text{hetero}} = X^T \Sigma^{-1} X$$

6.2 Cramér–Rao Lower Bound

The CRLB states that no unbiased estimator can have a smaller covariance than $I(\beta)^{-1}$:

$$\text{Cov}(\tilde{\beta}) \succeq I(\beta)^{-1}$$

OLS achieves the CRLB (homoscedastic case)

From Task 2: $\text{Cov}(\hat{\beta}_{\text{OLS}}) = \sigma^2(X^T X)^{-1} = I(\beta)^{-1}$. OLS hits the bound exactly. Numerically:

Table 8. Homoscedastic CRLB verification ($B = 2,000$). Max $|\text{Cov}(\text{OLS}) - \text{CRLB}| = 0.00$ (analytical identity).

Parameter	Empirical Var	CRLB	Ratio
$\hat{\beta}_0$	0.007916	0.008031	0.986
$\hat{\beta}_1$	0.000244	0.000244	0.999
$\hat{\beta}_2$	0.000303	0.000309	0.982

Ratios ≈ 1.0 confirm OLS is MVUE. The slight undershoot (< 1) is normal Monte Carlo variance with $B = 2,000$; the true ratio converges to 1.0 as $B \rightarrow \infty$.

WLS achieves the CRLB (heteroscedastic case)

For the heteroscedastic model: $\text{Cov}(\hat{\beta}_{\text{WLS}}) = (X^T \Sigma^{-1} X)^{-1} = I(\beta)^{-1}$. WLS hits its bound exactly. OLS does not:

Table 9. Heteroscedastic CRLB verification. WLS achieves the bound; OLS does not.

Parameter	OLS Emp. Var	WLS Emp. Var	WLS/CRLB	OLS/CRLB
$\hat{\beta}_0$	0.033652	0.010455	1.041	3.350
$\hat{\beta}_1$	0.001783	0.001295	1.034	1.425
$\hat{\beta}_2$	0.001340	0.000409	0.961	3.151

CRLB Summary

Homoscedastic: $\text{Cov}(\hat{\beta}_{\text{OLS}}) = \sigma^2(X^T X)^{-1} = I(\beta)^{-1}$ exactly — OLS is MVUE.

Heteroscedastic: $\text{Cov}(\hat{\beta}_{\text{WLS}}) = (X^T \Sigma^{-1} X)^{-1} = I(\beta)^{-1}$ exactly — WLS is MVUE.

OLS under heteroscedasticity exceeds the CRLB by 1.4–3.4 $\times$, quantifying the exact efficiency loss from model misspecification.

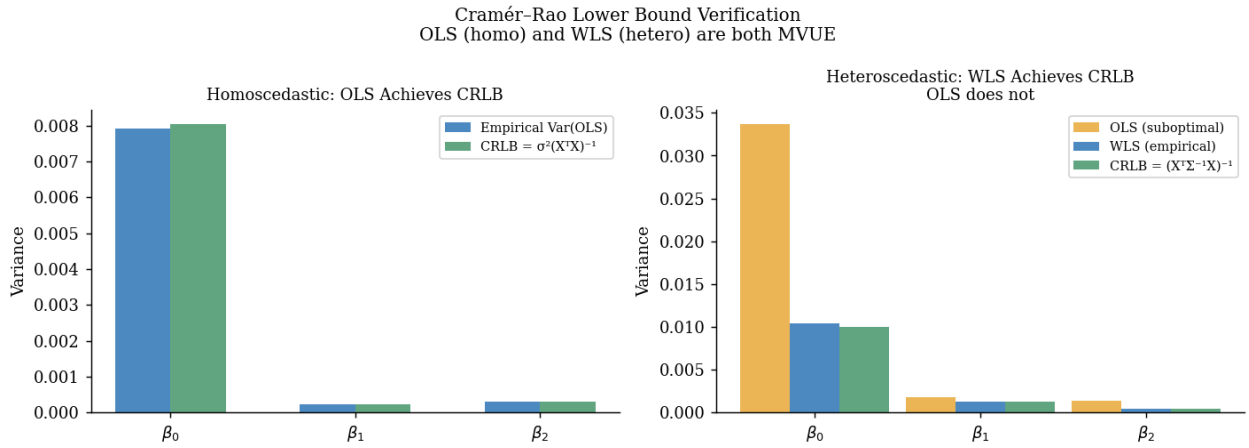


Figure 9. *Left:* OLS empirical variance vs. CRLB under homoscedastic noise — bars are identical, confirming MVUE. *Right:* OLS, WLS, and CRLB under heteroscedastic noise — WLS matches the CRLB; OLS exceeds it by 1.4–3.4 $\times$.

The OLS/CRLB ratios (3.35, 1.43, 3.15) are the exact efficiency penalties from Table 3: the gap between OLS and the theoretical minimum is not a modelling artifact but a provable, quantified consequence of misspecifying the noise structure.

6.3 Sufficient Statistics and the Rao–Blackwell Theorem

By the Fisher–Neyman Factorisation Theorem, the homoscedastic likelihood factors through $T(y) = (X^T y, \|y\|^2)$, making it a sufficient statistic for (β, σ^2) . Since $\hat{\beta}_{\text{OLS}} = (X^T X)^{-1} X^T y$ is a one-to-one function of $X^T y$, it is itself sufficient for β .

The **Rao–Blackwell theorem** states that any estimator that is a function of a sufficient statistic is at least as efficient as one that is not. Since OLS is a function of the complete sufficient statistic $X^T y$ and is unbiased, the **Lehmann–Scheffé theorem** guarantees it is the *unique* MVUE: no other unbiased estimator — linear or nonlinear — can achieve lower variance under Gaussian noise. This provides a second, complementary proof of MVUE status alongside the CRLB argument above.

Furthermore, $\|y\|^2$ together with $X^T y$ yields the residual sum of squares $\text{RSS} = \|y\|^2 - \hat{\beta}^T X^T y$,

from which the unbiased variance estimator $\hat{\sigma}^2 = \text{RSS}/(n - p)$ is derived — confirming that the sufficient statistic compresses all of y into $p + 1$ numbers without any loss of information about (β, σ^2) .

End of Report